

Data and text mining

DeepCDA: deep cross-domain compound–protein affinity prediction through LSTM and convolutional neural networks

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Abstract

Motivation: An essential part of drug discovery is the accurate prediction of the binding affinity of new compound–protein pairs. Most of the standard computational methods assume that compounds or proteins of the test data are observed during the training phase. However, in real-world situations, the test and training data are sampled from different domains with different distributions. To cope with this challenge, we propose a deep learning-based approach that consists of three steps. In the first step, the training encoder network learns a novel representation of compounds and proteins. To this end, we combine convolutional layers and long-short-term memory layers so that the occurrence patterns of local substructures through a protein and a compound sequence are learned. Also, to encode the interaction strength of the protein and compound substructures, we propose a two-sided attention mechanism. In the second phase, to deal with the different distributions of the training and test domains, a feature encoder network is learned for the test domain by utilizing an adversarial domain adaptation approach. In the third phase, the learned test encoder network is applied to new compound–protein pairs to predict their binding affinity.

Results: To evaluate the proposed approach, we applied it to KIBA, Davis and BindingDB datasets. The results show that the proposed method learns a more reliable model for the test domain in more challenging situations.

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Availability and implementation: <https://github.com/LBBSof/DeepCDA>.

1 Introduction

Compound–protein interaction (CPI) prediction, which determines the binding affinity of the interaction between a compound (drug candidate) and a target protein, plays a vital role in the drug discovery process (Chen *et al.*, 2016; Wen *et al.*, 2017). Experimentally measuring of compound–protein binding affinity is costly and time-consuming (Tian *et al.*, 2016; Tsubaki *et al.*, 2019). Hence, computational based models are getting more attention in CPI prediction (Masoudi-Sobhanzadeh *et al.*, 2019; Mousavian *et al.*, 2016). Pahikkala *et al.* (2015) explored the current methods in CPI prediction and found four factors that could impact the prediction results. These include (i) problem formulation (binary classification or

regression), (ii) evaluation datasets, (iii) evaluation procedure (simple or nested cross-validation) and (iv) the experimental setting. Also, they put forward some guidelines to formulate the model and evaluate it realistically. He *et al.* (2017) proposed a model called SimBoost. In their approach, the properties of drugs, targets and drug–target pairs are simultaneously fed into a model as input. Then the gradient boosting machine model is utilized for training the model. Moreover, they proposed SimBoostQuant in which the interval confidence is predicted for each compound–protein pair. In most classic machine learning approaches, the hand-crafted features are used. However, in recent years, it is shown that integrating feature learning capabilities into machine learning-based models improves prediction performance. Recently, deep learning-based models have

gotten more attention in many research areas such as drug discovery (Abbasi et al., 2019; Gawehn et al., 2016; Preuer et al., 2019), computer vision (Dai et al., 2016; Ouyang et al., 2015), text mining (Bian et al., 2014), bioinformatics (Li et al., 2019; Tang et al., 2019) and medical image processing (Shen et al., 2017; Xu et al., 2014). Deep learning-based models learn hierarchical representations of input data (LeCun et al., 2015). Kearnes et al. (2016) introduced an approach to automatically learn the molecule's features (molecular fingerprint), which is called graph convolutional network (GCN). GCN takes an undirected graph as an input in which each node corresponds to an atom, and each edge corresponds to a bond. GCN learns local substructures by combining information about each atom and its neighbors. Hirohara et al. (2018) used a convolutional neural network (CNN) to learn molecule representation. In this case, the SMILES notation of the molecule is given to the network as input to feed into the convolutional layers. Öztürk et al. (2018) introduced a model which is called DeepDTA. The sequence information of both proteins and compounds are fed into two different CNN networks. Then, the obtained feature vectors for proteins and compounds are concatenated and fed into three fully connected layers to predict the binding affinity. Tsubaki et al. (2019) used a GCN to learn compound features and used CNN to learn protein sequence features. Also, they utilized an attention mechanism to compute the attention coefficients, which considers the interaction strength of a molecule and a protein subsequence. Finally, the protein's feature vector is obtained using the weighted sum of protein subsequence's features along with attention coefficients. Karimi et al. (2019) utilized a seq2seq (Sutskever et al., 2014) auto-encoder model to learn protein and compound representation in an unsupervised manner. Next, the output of the learned encoder is fed into an attention layer, and the output is given to a one-dimensional (1D)-convolutional layer. The outputs of proteins convolutional layer and compounds convolutional layer are concatenated and fed into the fully connected layers.

One of the main challenges that we face during the CPI prediction is generalizing the model and finding the binding affinity of a new unseen compound–protein pair. There are four different experimental settings (Pahikkala et al., 2015) as the following: (i) warm setting: if both compound and protein of unseen pairs are seen in the training samples (widely used experimental setting), (ii) cold target setting: protein targets are not observed in the training set, (iii) cold drug setting: drug compounds are not observed in the training set and (iv) the case where both compounds and protein targets are not observed during the training phase that is the most challenging setting. The first three settings have gotten more attention than the last ones. Also, in most designed experiments, training and test sets are obtained from the same dataset. Whereas, in the real-world applications, the approach is applied to a test set that might be completely different from the training set. In other words, the training and test

sets might have completely different marginal distributions. In this study, we propose an approach to tackle this issue. The main goal is to learn a discriminative representation for the test set. To this end, we utilize the domain adaptation technique from transfer learning. In domain adaptation, the training and test sets are projected into a new space such that they have the same marginal distribution. Recently, domain adaptation has gotten more attention in many research areas like computer vision (Razzaghi, 2019; Razzaghi et al., 2019), molecule property prediction (Abbasi et al., 2019) and medical image processing (Mahmood et al., 2018).

The overall scheme of the proposed approach is shown in Figure 1. In the first step, we learn a feature encoder network and a predictor for the training domain using labeled training data (Fig. 1a). Next, a feature encoder network for the test domain is learned using training data and unlabeled test data (Fig. 1b). This step is done by utilizing an adversarial discriminative domain adaptation (ADDA) approach. Finally, the output of the test feature encoder network is fed into the training predictor to forecast the binding affinity for the new compound–protein pair (Fig. 1c).

The proposed feature encoder network utilizes a combination of CNN and long-short-term memory (LSTM) to describe both compound and protein. When CNN is applied to time-series data, it extracts highly informative features to encode local temporal patterns. In our approach, to capture the global temporal pattern, LSTM layers are added on top of the convolutional layers. In most approaches, to fuse the feature descriptor of compounds and proteins, they are simply concatenated (Karimi et al., 2019; Öztürk et al., 2018). Tsubaki et al. (2019), before concatenation, compute the importance weights of each subsequence in a protein and a compound using an attention mechanism. Then protein feature descriptor is modified based on these weights. Then, they are concatenated and are fed into the classification layers. In their approach, the aim of applying the attention mechanism is only to find the critical subsequences of protein, while finding the important local substructures of the compound is also essential in CPI prediction. In this paper, we propose a two-sided attention mechanism that encodes the mutual interaction of protein subsequences with compound substructures. The attention coefficient is computed between each of the compound and protein substructure pair that represents the binding strength between them. The feature encoder of DeepCDA is basically built on DeepDTA (Öztürk et al., 2018) by adding LSTMs and the attention mechanism. Then, the modified compound–protein feature descriptor is fed into the classification layers to predict the binding affinity. To evaluate the proposed approach, the test feature encoder network is learned using a transfer learning-based approach. In this case, situations with different distributions between tests and training domains are handled.

To sum up, the main contributions of this paper, which make it different from the other approaches, are as follows:

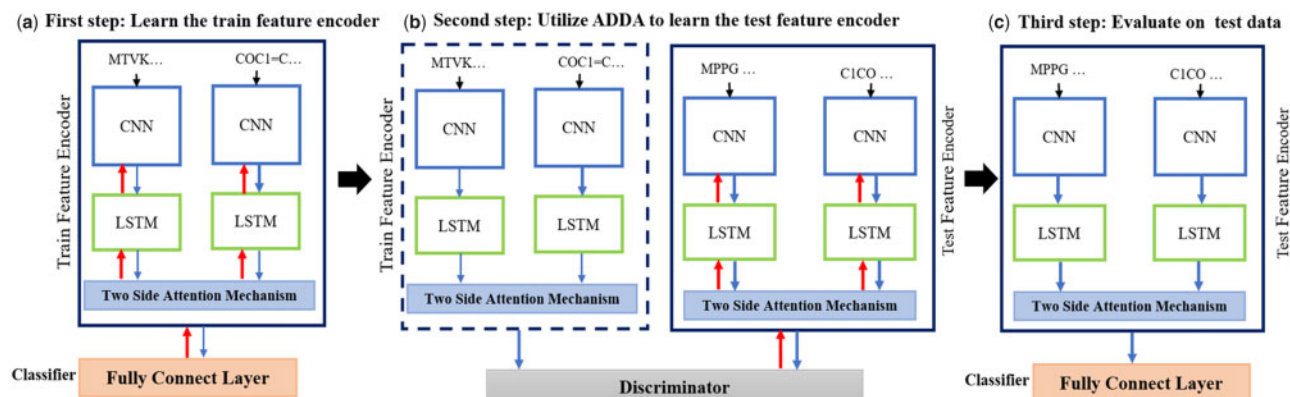


Fig. 1. The overall scheme of the proposed approach. (a) Learning the training feature encoder network using labeled training data (b) learning the test feature encoder network using unlabeled training and test data by utilizing adversarial domain adaptation approach (c) in the test phase, the new compound–protein pair is fed into the test feature encoder network and the output is fed into training prediction layer to forecast its binding affinity. The blue arrows denote the forward pass and the red arrows indicate the backward pass. (Color version of this figure is available at *Bioinformatics* online.)

1. Improving the generalizability of the model by utilizing the domain adaptation technique.
2. Proposing a combination of CNN and LSTM to get a better representation of protein and compound.
3. Proposing a two-sided attention mechanism that encodes the binding strength between each protein substructure and compound substructure pair.

2 Problem formulation

In CPI prediction, there are pairs of compounds and proteins in the training set. Let $X^{Train} = \{(c^{(i)}, p^{(i)})\}_{i=1}^S$ denote the set of training samples where c and p denote the compound and protein sequences, respectively, and N denotes the number of pairs that are available in the training phase. Also, the corresponding labels of each pair are denoted by $Y^{Train} = \{y^{(i)}\}_{i=1}^N$ which can be either a binary binding affinity (e.g. active or inactive) or a quantitative binding affinity (e.g. IC_{50} , K_d , K_i). Let C and T represent the compound space and protein space, respectively. Hence, we have $X^{Train} \subset C \times T$. The goal is to predict the binding affinity for the previously unseen pairs of compounds and proteins which are shown by X^{Test} . Up to now, in all approaches, the previously unseen pairs and training pairs are sampled from the same dataset. In this paper, we introduce a new setting in which the previously unseen pairs and training pairs are sampled from two different datasets. Therefore, the proposed approach is designed such that it generalizes the knowledge to predict the previously unseen pairs from a different distribution. To this end, we utilize the domain adaptation technique. In the following, we discuss the detail of the proposed approach.

3 The feature encoder network

In this step, the proposed architecture for the training feature encoder network is given. The overall architecture of this network is shown in Figure 2. The raw protein sequences and the compound SMILES strings are fed into the model as inputs. In the first step, these raw sequences are fed into encoding layers. Then, the outputs of the encoding layers are fed into the subsequent layers. In this paper, to encode the compound and protein sequence, a combination of CNN and LSTM is utilized.

Compound and protein sequences can be represented as a time series; hence, a recurrent network such as LSTM is suitable to encode the sequence. However, when the sequence is long, the training

of LSTM is hard due to the difficulty of encoding long-range dependencies. Hence, in the proposed approach, at first, the protein sequence is fed into CNN to extract its mid-level features in a hierarchical manner. Then, it is fed into an LSTM to encode the sequence's dependency. In the following, at first, we describe a vector space embedding of the protein sequence and compound sequence, then the architecture of CNN and LSTM is explained in detail. Up to now, CNN-LSTM combination is successfully applied in many research areas (Liu *et al.*, 2018; Wigington *et al.*, 2017; Wu *et al.*, 2018).

3.1 Embedding layer

Let $p = p_1, p_2, \dots, p_S$ denote the protein sequence, where p_i is the i -th amino acid. We use the n -gram embedding technique to obtain a vector representation of each protein. Since there are 20 kinds of amino acids; the total number of n -gram words is 20^n . Hence, generating a word for each n -gram is unfeasible. To this end, to incorporate the n -gram concept in the representation, each protein sequence is shown by:

$$p = (p_1, \dots, p_n), (p_2, \dots, p_{n+1}), \dots, (p_{P-n+1}, \dots, p_P). \quad (1)$$

As it is shown, the protein sequence consists of an overlapping n -gram amino acid. In this study, n is set to 3, and the maximum length of a protein sequence is considered to be 1000. To have fixed size lengths, proteins with lower length are padded with zeros. Then, the protein sequence is embedded into $p_e \in R^{P \times V_p}$ where V_p denotes the size of the amino-acid vocabulary. Each row of the matrix p_e denotes the embedding vector of n -gram amino acid.

A compound molecule is represented by an SMILES sequence (Weininger, 1988), which consists of a sequence of characters. Let $c = c_1, c_2, \dots, c_C$ denote a compound where c_i is an atom or a structure indicator and C is the length of the compound. In this study, the maximum length of the compound is set to 100. To have fixed size lengths, compounds with lower length are padded with zeros. The compound sequence is fed into the character embedding layer, thus transformed into a matrix. Hence, the output of the embedding layer is shown by $c_e \in R^{C \times V_c}$, where V_c denotes the size of SMILES vocabulary.

3.2 CNN

CNN is an architecture that consists of two common layers: (i) a convolutional layer and (ii) a pooling layer. The convolutional layer is a major building block of CNN, which contains a set of learnable filters where each filter is convolved with the input of the layer to encode the local knowledge of the small receptive field. The convolutional layer is commonly followed by the pooling layer. The pooling layer down-samples the output of the previous layer and does not have learning parameters. As a result, the receptive field of the subsequent convolutional layer is enlarged.

In the proposed approach, two different CNN architectures are used: one for the protein-embedded matrix and one for the compound molecule-embedded matrix. The output of protein CNN-block and compound CNN-block are, respectively, shown by $O_p^{cm} \in R^{l_p \times d_p}$ and $O_c^{cm} \in R^{l_c \times d_c}$ where d_p (d_c) denotes the number of feature maps and l_p (l_c) denote the length of the protein (compound) feature maps, respectively. Each feature map shows the strength of the existing local temporal pattern in each location. The learned local patterns (filters weights of the convolutional layer) of protein CNN represent the local substructure of the protein, which is useful in interaction prediction. These learned local substructures are called protein fragments. Also, the learned local structures of the compound are called compound fragments. The architecture of protein and compound CNN is shown in Figure 2.

CNN can only learn the local features of the sequence and cannot encode the long-range temporal inherent dependencies in the sequence. To overcome this issue, the output of CNN is fed into an LSTM layer. In this paper, the architecture of the CNN-block contains three consecutive 1D-convolutional layers. The second and the third convolutional layer have two to three times more filters than the first layer.

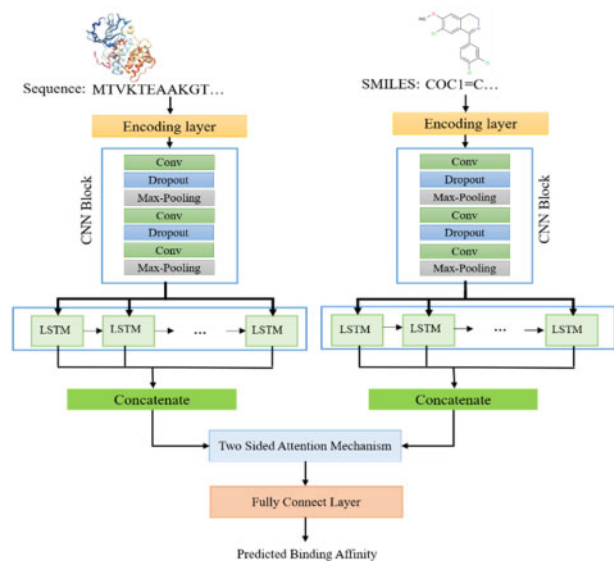


Fig. 2. The overall view of the feature encoder network

3.3 LSTM

LSTM is a recurrent neural network that takes a time-series sequence as an input and encodes the knowledge of the sequence (Hochreiter and Schmidhuber, 1997). Each feature map of the protein (compound) CNN shows the responses of a filter over the protein (compound) sequence. In this paper, to encode the order dependency of the learned local patterns in the sequence, we propose to feed each feature map into an LSTM layer. Figure 2 shows the architecture of the combination of the CNN and LSTM layers. As it is shown, each feature vector is fed into an LSTM layer. The output of each LSTM cell encodes the short-term and long-term dependencies observed up to that cell's input. The outputs of the LSTM layers are fused using concatenation to obtain the final feature vector. Therefore, the output of the concatenation layer for protein (compound) is shown by $O_p^{lstm} \in R^{l_p \times e}$ ($O_c^{lstm} \in R^{l_c \times e}$), where the parameter e shows the dimension of the embedding space of LSTM layers. The convolution layers try to find the contiguous portions of proteins that are effective in binding affinity value prediction. However, protein binding pockets that provide suitable properties for binding ligands are commonly formed by distant portions of the sequence. Hence, we have decided to utilize CNN-LSTM architecture, since CNN encodes a local range of input and LSTM encode long-range dependencies. In this paper, the number of hidden units of the LSTM layer is set to the number of filters of the last convolutional layers.

3.4 Two-sided attention mechanism

The attention mechanism helps us find out how different critical parts of the input in predicting the output are important. The attention mechanism is first introduced by Bahdanau et al. (2015). It assigns attention weight to each local part of the input, which plays an essential role in output prediction. Attention weights allow better visualization, which leads to more interpretable models. Recently, some approaches utilized the one-side attention mechanism in drug discovery. One of the contributions of this paper is the proposed two-sided attention mechanism. The aim of the two-sided attention mechanism is to generate the binding map, which weights the strength of each interaction between compound and protein fragments. Given $O_p^{lstm} = \{o_p^1, \dots, o_p^{l_p}\}$ as protein feature matrix and $O_c^{lstm} = \{o_c^1, \dots, o_c^{l_c}\}$ as a compound feature matrix, l_c and l_p denote the number of compound and protein fragments. At first, the mean feature descriptor over the feature map for compound and protein are computed as follows:

$$\hat{o}_p = \frac{1}{d_p} \sum_{i=1}^{l_p} o_p^i, \quad \hat{o}_c = \frac{1}{d_c} \sum_{i=1}^{l_c} o_c^i. \quad (2)$$

Then, the attention coefficient, which denotes the binding strength of each compound and protein fragments, is calculated as follows:

$$\alpha(i, j) = \sigma((o_p^i)^T W o_c^j) \sigma((\hat{o}_p)^T W \hat{o}_c). \quad (3)$$

The first part of Equation (3) computes the binding strength of i -th protein fragment and j -th compound fragment using their dot product, and the second part computes the binding strength between the mean feature vector of the compound and the mean feature vector of protein. Therefore, $\alpha(i, j)$ gets a higher value if both terms of Equation (3) get a higher value. Then, the weighted sum of compound and protein fragment descriptor is computed using the two-sided attention coefficient:

$$F = \sum_{i,j} \alpha(i, j) \text{concat} \left(o_p^i, o_c^j \right). \quad (4)$$

This vector is fed into classification layers. The parameters of the training feature encoder network are learned by optimizing the mean squared error (MSE) loss function, which is defined as follows:

$$\sum_{x^{(i)} \in X^{train}, x^{(i)} = (p^{(i)}, c^{(i)})} \left(N^{train}(x^{train}) - y^{(i)} \right)^2, \quad (5)$$

where N^{Train} denote the training feature encoder network.

4 Adversarial domain adaptation

In this step, the domain adaptation between the training feature encoder network and the test feature encoder network is performed. If the training and the test domains have different marginal distributions, the domain adaptation technique is utilized to map these domains into a new feature space where they have the same distribution. In recent years, ADDA techniques have gotten more attention (Chadha and Andreopoulos, 2018; Chen et al., 2018; Tzeng et al., 2017). In ADDA, an embedding subspace is learned using a minimax game strategy.

As it is shown in Figure 1b, there are three sub-networks: (i) the training feature encoder network, (ii) the test feature encoder network and (iii) the discriminator layers (consists of some fully connected layers). Let N^{Test} denote the test feature encoder network and D be the discriminator network. The test feature encoder network (N^{Test}) is copied from the training feature encoder network (N^{Train}). In this case, there are two loss functions: the discriminator loss and the adversarial loss. Discriminator loss learns to distinguish the training domain samples from the test domain samples which are defined as follows:

$$\begin{aligned} L_{adv,D}(N^{Train}, N^{Test}, D; X^{Train}, X^{Test}) \\ = - \sum_{x^{train} \sim p(x^{train})} \log(D(N^{Train}(x^{train}))) p(x^{train}) \\ - E_{x^{test} \sim X^{Test}} [1 - \log(D(N^{Test}(x^{test})))], \end{aligned} \quad (6)$$

where $p(x^{train})$ shows the marginal distribution of the training samples which is defined as follows:

$$p(x^{train}) \propto \exp \left(- \frac{\min_i \|\alpha^{train} - \alpha^i\|}{2\varepsilon} \right), \quad (7)$$

where α^{train} and α^i denote the attention maps of the training data and the i -th test data. This distribution assigns a higher probability upon the training samples that have more similar binding maps with test samples. During the minimization of $L_{adv,D}$, the training and test feature encoder networks are fixed, and only the discriminator layers are updated. Adversarial loss tries to learn a transferable shared feature space for the test domain. To do so, it fools the discriminator to make the test domain look like the training domain. Hence, adversarial loss is defined as follows:

$$L_{adv,G}(N^{Test}, D; X^{Test}) = -E_{x^{test} \sim X^{Test}} [\log(D(N^{Test}(x^{test})))]. \quad (8)$$

It should be noted that during the minimization of $L_{adv,G}$, the training feature encoder network and the discriminator layers are fixed, and only the test feature encoder network is updated. The loss functions $L_{adv,D}$ and $L_{adv,G}$ are optimized iteratively until they converge. It is expected that in the convergence state (which is a Nash-equilibrium state) feature encoder of the test domain and the training domain share the same distributions (Mescheder, 2018). In other words, the domain adaptation techniques try to learn more discriminative feature encoder for the test domain when the training and test domains have different marginal distributions.

5 Inference step

In this step, the binding affinity for an unseen pair of compound-protein is predicted. In the previous step, the feature encoder network for the test domain is learned. By feeding the previously unseen pair of compound-protein into this network, the final feature vector is provided as output. Finally, we can obtain the predicted

binding affinity by applying the training predictor layers on this feature vector.

6 Model validation

In this section, the proposed approach is evaluated on three common datasets: KIBA dataset (Tang *et al.*, 2014), Davis dataset (Davis *et al.*, 2011) and BindingDB dataset (Liu *et al.*, 2007). As it is stated in this paper, three different contributions are proposed. Hence, we conduct two different settings for experiments to evaluate these contributions. In the first setting, we design some experiments to examine the effectiveness of the proposed feature encoder network. The second setting of the experiment assesses the generalizability of the proposed approach to transfer knowledge between two different datasets.

To compare the proposed approach with the current state-of-the-art approaches, we choose KronRLS (Pahikkala *et al.*, 2015), SimBoost (He *et al.*, 2017), DeepAffinity (Karimi *et al.*, 2019) and DeepDTA (Öztürk *et al.*, 2018) as baseline approaches. It should be noted that the results of base approaches are taken directly from their original publications, except it is clearly mentioned that it is trained by us.

To evaluate the performance of the proposed approach on KIBA and Davis datasets, we utilize the nested cross-validation approach similar to He *et al.* (2017) and Öztürk *et al.* (2018). To this end, the dataset is randomly divided into six equal parts. One part is chosen as a test set, and the remaining parts are utilized for training via five-fold cross-validation. Hence, all reported measures are the average of the five results. It should be noted that to have a fair comparison, the proposed approach and all base approaches use a similar test set and training set. Also, to evaluate the performance of the proposed approach on the BindingDB dataset, the partition of the training and test sets is similar to Karimi *et al.* (2019).

6.1 Datasets

The KIBA dataset combines kinase inhibitor bioactivities from varying sources such as IC_{50} , K_i and K_d into KIBA score (Tang *et al.*, 2014). The original KIBA dataset contains a matrix of 467 proteins and 52 498 compounds, with 246 088 interaction. He *et al.* (2017) filtered the original KIBA dataset by removing all proteins and compounds with <10 interactions.

The Davis dataset contains interactions of 442 unique proteins and 68 unique compounds, measured by the K_d value. It reports respective dissociation constant (K_d) values of the kinase protein family and the relevant inhibitors. Similar to He *et al.* (2017), the K_d measure is transformed into log space (pK_d) as follows:

$$pK_d = -\log_{10}\left(\frac{K_d}{1e9}\right). \quad (9)$$

The BindingDB is a public and web-accessible dataset. In this study, we use K_f -labeled samples. The four classes of proteins are entirely excluded from the training set: nuclear estrogen receptors with 516 samples, ion channels with 8101 samples, receptor tyrosine kinases with 3355 samples and G-protein-coupled receptors with 77 994 samples to test the generalizability of the approach. The remaining ones, similar to Karimi *et al.* (2019), are divided into the

training set and the test set, which, respectively, contain 101 134 and 43 391 samples.

Table 1 lists the details of datasets, including the number of unique proteins, number of unique compounds, number of interactions between protein and compound. Also, the maximum length and the average length of the protein sequence and compound are given.

In this study, the predicted continuous values are also converted into binary values by applying thresholds. Similar to He *et al.* (2017), the selected thresholds for the Davis, KIBA and BindingDB datasets are set to 7, 12.1 and 7.6, respectively. Hence, in Table 1, the number of active and inactive compounds for each dataset is also reported. As it is shown, all datasets are imbalanced and have skewed distribution. The imbalanced ratios for Davis, KIBA and BindingDB datasets, respectively, are 0.082, 0.192 and 0.349. Thus, the choice of appropriate evaluation measures is crucial.

6.2 Evaluation metrics

In this study, five evaluation measures are used: (i) concordance index (CI), (ii) MSE, (iii) r_m^2 index, (iv) the Pearson correlation coefficient and (v) area under the precision-recall curve (AUPR). The four first measures are used to evaluate the model with continuous output and the last one is utilized to evaluate the model with binary output.

CI is a model assessment measure that is introduced by Gönen and Heller (2005). It measures the probability of the concordance between the ground truth value and the predicted value. Let δ_i and b_i , respectively, denote the ground truth value and predicted value of the i -th sample. Hence, this measure is defined as follows:

$$CI = \frac{1}{Z} \sum_{i,j} \sigma(\delta_i > \delta_j) b(b_i - b_j), \quad (10)$$

where Z denotes the normalization constant and σ is a step function which returns one if condition statement is satisfied otherwise it returns zero. Also, $b(x)$ is defined as follows:

$$b(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0.5 & \text{if } x = 0 \\ 0 & \text{if } x < 0. \end{cases} \quad (11)$$

The range of value for CI is between 0 and 1 where the value of one denotes the best result. The Pearson correlation (R) is a metric that measures the linear correlation between two variables. This measure is varied between -1 and $+1$. If the two variables are completely correlated it takes $+1$ and if they are reversely correlated then it takes -1 . If there is no correlation between variables then it takes zero.

MSE is a well-defined measure which is defined as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (\delta_i - b_i)^2. \quad (12)$$

Regression toward the mean (r_m^2 index) is a modified, squared correlation coefficient measure that reflects the external predictive potential of a model. This measure is defined as:

Table 1. Details of datasets contain unique proteins, unique compounds and interactions between protein and compounds

	No. proteins	No. compounds	Interaction			Protein		Compound	
			No. interaction	No. active	No. inactive	Max length	Avg length	Max length	Avg length
Davis	442	68	30 056	2457	27 554	2549	788	103	64
KIBA	229	2111	118 254	22 729	93 426	4128	728	590	58
BindingDB	81 417	79 536	234 491	81 743	152 748	1485	239	101	28

Note: Also, the maximum length and the average length of the protein sequence and compound are given.

$$r_m^2 = r^2 \times \left(1 - \sqrt{r^2 - r_0^2}\right), \quad (13)$$

where r_0^2 is a squared correlation coefficient with zero intercepts. An acceptable model should have an r_m^2 index value greater than 0.5. Roy et al. (2013) showed that regression toward the mean is extensively used in regression-based QSAR models.

The AUPR assesses a binary model by averaging the precision across all recall values. As it is mentioned, all datasets are imbalanced and have skewed distribution. Hence, it leads us to choose AUPR because the precision–recall curve is more appropriate than the receiver operating characteristic (ROC) curve for imbalanced data. AUPR is suitable for tasks in which there is a significant skew in the class distribution. In this study, the predicted continuous values are converted into binary values by applying thresholds introduced in He et al. (2017).

Also, the hyperparameters optimization is done for the number of the filters (same for proteins and compounds) search over [16, 32, 64, 128, 256], the length of the filter size for compounds, the length of the filter size for proteins, the learning rate for training feature encoder network search over [0.1, 0.01, 0.001, 0.0001], the learning rate for test feature encoder network search over [0.001, 0.0005, 0.0001, 0.00005, 0.00001] and the learning rate for discriminator layers search over [0.001, 0.0005, 0.0001, 0.00005, 0.00001].

In this paper, we run the program on the computer of Intel(R) Core(TM) i7-7700HQ CPU, NVIDIA GeForce GTX 1070 8Gb GDDR5 and 32G DDR4 RAM. We implemented our method using python 3.6, TensorFlow and Keras. Each iteration takes around 0.01 s during training with a minibatch size of 256.

6.3 Results

The results obtained from applying our approach to the KIBA dataset are reported in Table 2. Also, our result is compared with the base approaches. As it is shown, in all of the evaluation measures, our approach outperforms the base approaches. Our approach gets a 0.026 increase in CI index and the MSE value of our model is 0.018 lower than that of the DeepDTA. The evaluation measure of AUPR is provided to assess the ability of the model in binary prediction. Our approach achieves around 0.024 improvement in AUPR compared to the best comparable approach. To show the importance of the selected architecture, we report the obtained results for two other versions of the proposed approach: (i) DeepCDA (only LSTM): in this case, only LSTM layers are utilized as feature extraction layers of the proposed approach and (ii) DeepCDA + w.o.

Attention: in this case, the attention mechanism is not utilized in the proposed method. It should be noted that DeepDTA only uses convolutional layers to extract features. The obtained results on these two versions are shown in Table 2. As it is shown, by comparing DeepCDA + w.o. Attention with DeepCDA, it is concluded that utilizing attention mechanism achieves a 0.014 improvement in the CI index.

To statistically evaluate the significant improvement of our method, the paired *t*-test is utilized at a significant level of 0.05. This test shows that our approach outperforms DeepDTA with a *P*-value of lower than 0.0001. Also, Öztürk et al. (2018) showed that DeepDTA outperforms SimBoost and KronRLS with a *P*-value of around 0.0001 for both.

Table 3 shows the achieved results on the Davis dataset. In this case, our approach gets a better result in all measures compared to the baseline approaches. Our approach gets 0.013, 0.019 and 0.025 improvements in CI index, r_m^2 and AUPR over DeepDTA, respectively. Also, the MSE measure is 0.248 for the Davis dataset. In this dataset, the obtained results on both DeepCDA (only LSTM) and DeepCDA + w.o. Attention are also reported. When only LSTM layers are used as feature extraction layers, it gets a 0.014 decrement in CI index compared to DeepDTA. Also, incorporating attention layers gets 0.005 improvements in CI index in comparison with DeepDTA.

The paired *t*-test shows that our approach outperforms DeepDTA with a *P*-value of around 0.001. Also, it is shown that DeepDTA outperforms KronRLS and simBoost with a *P*-value of 0.0001 for both (Öztürk et al., 2018).

The obtained results on the BindingDB dataset are shown in Table 4. As it is shown, the full proposed approach gets a better Pearson correlation coefficient (*R*) than DeepAffinity and DeepDTA. Moreover, Table 4 shows that the combination of CNN and LSTM, as a feature encoder network, performs better than only CNN (DeepDTA) and only LSTM [DeepCDA (only LSTM)] as feature encoders. Also, utilizing the domain adaptation technique in DeepCDA leads to an improvement in all metrics over DeepCDA + w.o. Attention. The paired *t*-test shows that our approach outperforms DeepDTA with a *P*-value of around 0.01.

6.4 Transferability

The generalization of the CPI prediction method is important in drug discovery and development. In this section, the generalization ability of the proposed approach in transferring knowledge between

Table 2. Comparison of our approach and base comparable approaches on the KIBA dataset

	CI ± std	<i>R</i>	MSE	r_m^2 ± std	AUPR ± std
KronRLS	0.782 ± 0.0009	—	0.411	0.342 ± 0.001	0.635 ± 0.004
simBoost	0.836 ± 0.001	—	0.222	0.629 ± 0.007	0.760 ± 0.003
DeepDTA	0.863 ± 0.002	0.848	0.194	0.673 ± 0.009	0.788 ± 0.004
DeepCDA (only LSTM)	0.852 ± 0.012	0.817	0.214	0.633 ± 0.024	0.781 ± 0.017
DeepCDA + w.o. Attention	0.877 ± 0.017	0.855	0.189	0.678 ± 0.009	0.803 ± 0.007
Our approach (DeepCDA)	0.889 ± 0.002	0.855	0.176	0.682 ± 0.008	0.812 ± 0.005

Note: In this experiment, the *R*, CI, MSE, r_m^2 and AUPR scores are reported, and std stands for standard deviation.

Table 3. Comparison of our approach and base comparable approaches on Davis dataset

	CI ± std	<i>R</i>	MSE	r_m^2 ± std	AUPR ± std
KronRLS	0.871 ± 0.0008	—	0.379	0.407 ± 0.005	0.661 ± 0.010
simBoost	0.872 ± 0.002	—	0.282	0.644 ± 0.006	0.709 ± 0.008
DeepDTA	0.878 ± 0.004	0.846	0.261	0.630 ± 0.017	0.714 ± 0.010
DeepCDA (only LSTM)	0.864 ± 0.002	0.852	0.287	0.62 ± 0.021	0.71 ± 0.023
DeepCDA + w.o. Attention	0.883 ± 0.009	0.849	0.261	0.63 ± 0.017	0.70 ± 0.009
Our approach (DeepCDA)	0.891 ± 0.003	0.857	0.248	0.649 ± 0.009	0.739 ± 0.006

Note: In this experiment, the CI, *R*, MSE, r_m^2 and AUPR scores are reported, and std stands for standard deviation.

Table 4. Comparison of our approach and base comparable approaches on the BindingDB dataset

	CI $\pm$ std	R	MSE	$r_m^2 \pm$ std	AUPR $\pm$ std
DeepAffinity ^a	—	0.81	0.91	—	—
DeepDTA	0.812 $\pm$ 0.02	0.832	0.824	0.623 $\pm$ 0.02	0.443 $\pm$ 0.01
DeepCDA (only LSTM)	0.796 $\pm$ 0.03	0.781	1.021	0.574 $\pm$ 0.02	0.378 $\pm$ 0.02
DeepCDA + w.o. Attention	0.818 $\pm$ 0.02	0.826	0.813	0.625 $\pm$ 0.01	0.451 $\pm$ 0.09
Our approach (DeepCDA)	0.822 $\pm$ 0.01	0.844	0.808	0.631 $\pm$ 0.02	0.459 $\pm$ 0.03

Note: In this experiment, the CI, R, MSE, r_m^2 and AUPR scores are reported, and std stands for standard deviation.

^aReported from Karimi *et al.* (2019).

Table 5. Comparison of our approach and base comparable approaches on knowledge transfer from KIBA to Davis and BindingDB datasets

Method	KIBA $\rightarrow$ Davis		KIBA $\rightarrow$ BindingDB	
	CI $\pm$ std	AUPR $\pm$ std	CI $\pm$ std	AUPR $\pm$ std
DeepDTA	0.67 $\pm$ 0.04	0.33 $\pm$ 0.023	0.53 $\pm$ 0.02	0.33 $\pm$ 0.03
DeepDTA + w. DA	0.72 $\pm$ 0.04	0.59 $\pm$ 0.02	0.56 $\pm$ 0.01	0.37 $\pm$ 0.02
DeepCDA + w.o. DA	0.68 $\pm$ 0.03	0.41 $\pm$ 0.02	0.53 $\pm$ 0.04	0.34 $\pm$ 0.02
DeepCDA	0.74 $\pm$ 0.03	0.59 $\pm$ 0.01	0.57 $\pm$ 0.02	0.38 $\pm$ 0.01

Note: In this experiment, the CI and AUPR scores are reported, and std stands for standard deviation. The bold values indicate the better results obtained in the experiments.

Table 6. Comparison of our approach and base comparable approaches on knowledge transfer from Davis to KIBA and BindingDB datasets

Method	Davis $\rightarrow$ KIBA		Davis $\rightarrow$ BindingDB	
	CI $\pm$ std	AUPR $\pm$ std	CI $\pm$ std	AUPR $\pm$ std
DeepDTA	0.55 $\pm$ 0.01	0.30 $\pm$ 0.001	0.49 $\pm$ 0.02	0.10 $\pm$ 0.03
DeepDTA + w. DA	0.60 $\pm$ 0.04	0.52 $\pm$ 0.006	0.51 $\pm$ 0.02	0.11 $\pm$ 0.01
DeepCDA + w.o. DA	0.58 $\pm$ 0.01	0.39 $\pm$ 0.002	0.48 $\pm$ 0.03	0.11 $\pm$ 0.01
DeepCDA	0.64 $\pm$ 0.03	0.58 $\pm$ 0.001	0.52 $\pm$ 0.03	0.12 $\pm$ 0.01

Note: In this experiment, the CI and AUPR scores are reported, and std stands for standard deviation. The bold values indicate the better results obtained in the experiments.

two different datasets is evaluated. To this end, our approach is compared with three base approaches including DeepDTA, DeepAffinity and our approach without domain adaptation. Also, to show the power of domain adaptation on compound–protein affinity prediction, we apply our domain adaptation technique on the DeepDTA method which is abbreviated by DeepDTA + w. DA in Tables. It can demonstrate that properly applying domain adaptation techniques can lead to an increase in performance. It should be noted that since the binding affinities in KIBA, Davis and BindingDB datasets are measured from different sources, only CI and AUPR measures are utilized to compare the approaches.

In this experiment, three different settings are designed. In the first case, the KIBA dataset is considered as the training dataset, and Davis and BindingDB datasets are considered as the test sets. The obtained results are shown in Table 5. As it is shown, the KIBA dataset has more knowledge to transfer to the Davis dataset rather than the BindingDB dataset. The reason is both KIBA and Davis datasets comprise kinase proteins while BindingDB contains more diverse protein families. Also, as it is presented in Table 5, by applying ADDA on DeepDTA, we see an increase in both CI and AUPR metrics. In the second case, Davis dataset is used to train the model and then it is evaluated on KIBA and BindingDB datasets. As it is shown in Table 6, our approach gets a significant improvement in both CI and AUPR measures in Davis $\rightarrow$ KIBA. It should be noted that, in the test phase, our approach learns a new feature encoder for both compounds and proteins to have a more discriminative and better representation. Hence, in cases where the approach gets more

improvement, the approach learns the better representation of compounds and proteins. In Davis $\rightarrow$ KIBA, our full approach significantly improves both performance measures. Despite the low variety of compounds in Davis and the contrast high variety of compounds in the KIBA dataset, our approach successfully transfers the knowledge between datasets. However, in Davis $\rightarrow$ BindingDB, our approach cannot get a significant improvement. By comparing the results of KIBA $\rightarrow$ BindingDB and Davis $\rightarrow$ BindingDB, it is found out that KIBA has more power to transfer knowledge to BindingDB. The reason is that the KIBA dataset has more diverse interaction pairs than the Davis dataset. It should be noted that in this experiment, we download the DeepDTA code and run their code based on the setting of their hyperparameters. In their approach, the parameter search is done to find the best set of hyperparameters. In Davis $\rightarrow$ KIBA, Table 6 shows that DeepDTA + w. DA gets 0.05 and 0.22 improvements in CI and AUPR over DeepDTA, respectively. It denotes that the proposed domain adaptation technique is effective in transferring knowledge.

In the third setting, the training set of the BindingDB dataset is utilized as the training set to learn the model, and then it is applied to ER, ion channel, GPCR, kinase, KIBA and Davis datasets which all of them are completely different from BindingDB training dataset. It should be noted that BindingDB, KIBA and Davis binding affinity values are stated in different scores; hence, for KIBA and Davis datasets, only CI and AUPR measures are reported. Table 7 shows that, in the Pearson correlation coefficient (R), our approach gets better results than comparable approaches. However, in MSE,

Table 7. Comparison of our approach and base comparable approaches on knowledge transfer from BindingDB as a training dataset to different test datasets

Datasets	Metrics	Methods				
		DeepAffinity	DeepDTA	DeepDTA + w. DA	DeepCDA + w. DA	DeepCDA
ER	MSE	1.76	3.73	3.91	3.03	3.85
	R	0.09	0.004	0.04	0.05	0.10
	CI	—	0.50	0.52	0.51	0.53
	AUPR	—	0.10	0.11	0.12	0.12
	r_m^2	—	0.006	0.01	0.007	0.004
Ion channel	MSE	1.79	2.51	2.78	2.25	2.50
	R	0.23	0.31	0.31	0.32	0.31
	CI	—	0.60	0.61	0.60	0.60
	AUPR	—	0.19	0.20	0.20	0.22
	r_m^2	—	0.07	0.05	0.06	0.08
GPCR	MSE	1.50	2.15	2.59	2.04	2.67
	R	0.21	0.26	0.28	0.27	0.28
	CI	—	0.58	0.59	0.57	0.60
	AUPR	—	0.15	0.13	0.14	0.15
	r_m^2	—	0.05	0.05	0.02	0.06
Tyrosine kinase	MSE	2.10	3.94	4.08	3.98	3.14
	R	0.16	0.26	0.36	0.28	0.42
	CI	—	0.61	0.65	0.63	0.67
	AUPR	—	0.14	0.17	0.16	0.19
	r_m^2	—	0.04	0.09	0.04	0.13
KIBA	MSE	—	—	—	—	—
	R	—	—	—	—	—
	CI	—	0.53	0.53	0.54	0.52
	AUPR	—	0.41	0.42	0.39	0.44
	r_m^2	—	—	—	—	—
Davis	MSE	—	—	—	—	—
	R	—	—	—	—	—
	CI	—	0.53	0.56	0.54	0.58
	AUPR	—	0.19	0.20	0.20	0.21
	r_m^2	—	—	—	—	—

Note: In this experiment, CI, MSE, r_m^2 and AUPR are reported.

DeepAffinity gets the best results compared to DeepDTA and DeepCDA. Table 7 shows that applying the utilized domain adaptation technique in this paper on DeepDTA leads to an improvement in most of the measures over DeepDTA. In most cases, our approach gets improvements in CI, R and AUPR measures. However, in transferring knowledge from BindingDB to the KIBA dataset, the proposed approach cannot improve the CI measure and it is dropped. In this case, the negative transfer has occurred. This challenge will be considered in future work. In standard machine learning algorithms, it is assumed that the training and test sets are drawn from the same distribution. Transfer learning copes with this challenge by assuming that the training and test data might have drawn from different distributions. In some cases, like DeepAffinity, a technique called a warm restart is utilized in which the model is trained on a huge dataset and then it is refined on the ‘target’ dataset. The obtained results listed in Table 7 denote that utilizing domain adaptation technique outperforms DeepAffinity, which utilizes the warm restart technique, in most evaluation metrics except MSE.

6.5 Target selectivity of drugs

In this section, we assess how well the proposed approach predicts the target selectivity of certain drugs. Protein-tyrosine phosphatases (PTPs) family plays a significant role in metabolic regulation and mitogenic signal transduction processes. In this experiment, three compounds, such as PTP inhibitors, are chosen, which include 2-(oxaloamino) benzoic acid, 3-(carboxyformamido)thiophene-2-carboxylic acid, and 2-formamido-4,5,6,7-tetrahydrothieno [2,3-c]pyridine-3-carboxylic acid which their, respectively, PubChem CIDs are 444764, 44359299 and 90765696. Also, we choose five

Table 8. The pK_i value for the compound 1, compound 2 and compound 3 against PTP1B

	Compound 1	Compound 2	Compound 3
pK_i (against PTP1B)	4.63	4.25	6.69
ΔpK_i (the closest difference)	0.75	0.7	2.47

Note: The closes differences of pK_i against the other proteins of the PTP family are also given.

proteins in the PTP family that contain PTP1B, PTPRA, PTPRE, PTPRC and SHP1. It is shown that the selectivity of the mentioned compounds leads highly toward PTP1B rather than the other four proteins (Iversen et al., 2000). The pK_i value for the mentioned compounds against PTP1B and its difference compared to the closest protein of the PTP family are given in Table 8.

In this experiment, the results of DeepDTA and our approach without domain adaptation (DeepCDA + w.o. DA) are compared with the results of our approach. In Table 9, the predicted pK_i values for three approaches are shown. Our full approach can predict more selectivity of compound 2 and compound 3 for PTP1B protein. However, both DeepDTA and DeepCDA + w.o. DA predicts PTP1B selectivity correctly for only one compound. It should be noted that none of the five mentioned proteins and three mentioned compounds exist in the training set.

To explore the low efficiency of our approach in the prediction of the affinity of compound 1 in comparison to compound 2 and compound 3, the similarities between compounds are investigated.

To this end, we have utilized the OpenBabel descriptor (O'Boyle *et al.*, 2011). We find out that compound 1 is different from compound 2 and compound 3 in some different features including the number of hydrogen bond acceptors 1 (JoelLib) (HBA1), octanol/water partition coefficient ($\log P$), molar refractivity (MR) and molecular weight filter. For example, the $\log P$ states that how the tendency an organic compound has to be adsorbed in soil and living organism which its values for three compounds are, respectively, achieved 1.69, 0.54 and 0.48. Also, the MR values for three compounds are 60.51, 47.36 and 49.48, respectively. It is shown that these features are important in binding affinity prediction.

It should be considered that, generally, explainability and interpretability of machine learning is currently one of the major research areas. Hence, in some cases, the approach fails to provide an appropriate explanation for why the algorithm could not do well, because the explainability of the deep learning-based models is still abstract.

6.6 CPI visualization with attention weights

In this section, the main goal is to demonstrate that the attention mechanism could lead to more explainability in the deep learning-based model by providing which regions in a protein and drug compound have more effective roles for interactions between a drug compound and a protein. In this case, the more important regions

Table 9. Predicted pK_i value for selective three compounds against five proteins in the human PTP family

Protein	DeepDTA			DeepCDA + w.o. DA			DeepCDA		
	C.1	C.2	C.3	Co.1	C.2	C.3	C.1	C.2	C.3
PTP1B	5.49	5.79	5.84	5.49	5.83	5.78	5.61	5.93	6.12
PTPRA	5.70	5.60	5.54	5.66	5.64	5.46	5.70	5.13	5.93
PTPRE	5.86	5.89	5.78	5.83	5.77	5.66	5.23	5.62	5.01
PTPRC	5.66	5.39	5.74	5.63	5.56	5.88	4.99	5.37	5.28
SHP1	5.72	5.67	5.50	5.85	5.73	5.71	5.35	5.46	5.69

Note: The bold values indicate the better results obtained in the experiments.

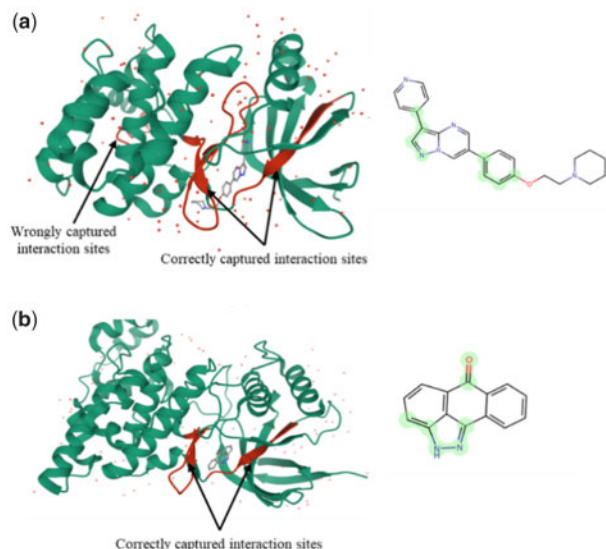


Fig. 3. The visualization of interaction sites in a protein and drug compound for (a) human AMP-activated protein kinase alpha 2 subunit kinase domain (T172D) complexed with compound C (PDB ID: 3AQV). The obtained residue numbers by our approach are 87–96, 143–167 and 211–219, and (b) JNK3 in complex with a dihydroanthrapyrazole inhibitor (PDB ID: 1PMV). The obtained residue numbers by our approach are 141–149 and 193–205

are highlighted by high-value attention weights. To visualize the interaction sites, at first, the attention weights are computed. Next, feature indexes which their corresponding attention values are local maximum and greater than the 80% of the maximum value are chosen as the interaction sites. Each feature sequence is associated with a receptive field on input. To compute the corresponding receptive field centers of the interaction sites, the introduced approach in Araujo *et al.* (2019) is utilized. Finally, the computed receptive field is represented in the input space. Figure 3 shows such regions that are highlighted for two different examples in which their three-dimensional (3D) interaction structures are known. Figure 3a shows the human AMP-activated protein kinase alpha 2 subunit kinase domain (T172D) complexed with compound C (PDB ID: 3AQV). The interaction is defined by a distance in the range of <4.0 Å. Based on the NCBI structure database, the residue count for 3AQV is 13. These residues include: 43, 45, 94–99, 103–104, 146, 156 and 164 (PDB residue number). The obtained residue numbers by our approach are 87–96, 143–167 and 211–219. These regions are shown by red colors on the 3D structure (colored cartoon representation). In Figure 3a, the important substructures of the corresponding compound of 3AQV are also shown. These substructures achieve maximum attention weights. As it is shown, two substructures are founded where the first one is a substructure similar to the Pyrazole group, and the second one is similar to a phenol group. Based on the information taken from the NCBI structure database, it is confirmed that the first substructure is a truly detected binding substructure. Figure 3b shows the structure of JNK3 in complex with a dihydroanthrapyrazole inhibitor (PDB ID: 1PMV). Based on the information taken from the NCBI structure database, the residue count for 1PMV is 11 which include: 70, 91, 146–152, 197 and 206 (PDB residue number). The obtained residue numbers by our approach are 141–149 and 193–205. Also, in Figure 3b, the significant substructures of 2,6-Dihydroanthra-1,9-CdPyrazol-6-One in complex with JNK3 are shown. As it is shown, two substructures are detected in which the first one is similar to an acetyl group and the second one is similar to the indazole ring. As it is shown in Figure 3a, two regions with higher attention weights have more overlap with the interaction sites. However, in Figure 3a, there is a region with higher attention weight, which is not an interaction site. In Figure 3b, our approach could assign higher attention weights to regions that have overlap with interaction sites. However, in both cases, some other interaction sites cannot get higher attention weight.

7 Discussion

We have shown that by integrating domain adaptation technique into the CPI prediction algorithm, we can achieve a more accurate prediction on new compound–protein pairs from a different domain. One of the main goals of CPI prediction is to predict a binding affinity value for a new compound–protein pair. To this end, we proposed a new architecture for CPI prediction which combines a convolutional network and LSTM layers into a unified framework to effectively encode the local and global temporal patterns. To fuse the compound and protein descriptors, a two-sided attention mechanism is proposed, which computes the binding strength between compound and protein subsequences. Since the test domain might differ from the training domain, we utilize the adversarial domain adaptation network to learn new feature encoder for the test domain.

To evaluate the proposed work, it is applied to three common datasets: KIBA, Davis and BindingDB datasets. The variety of proteins in Davis dataset is more than the KIBA dataset and in contrast, KIBA has a wide variety of compound types than the Davis dataset. The BindingDB dataset has a wide variety in both compound and protein types than the Davis and KIBA datasets.

One of the findings of the proposed approach is that if the test feature encoder network fine-tuned using a domain adaptation technique, then the knowledge transfer occurred between the training and test domains, lead to an improvement in performance. When Davis dataset is used as a training domain, and KIBA is selected as the test domain, the approaches without domain adaptation do not achieve acceptable results. However, our approach can build a

better predictive model for the test domain. It is observed that the approaches without domain adaptation, in KIBA → Davis, get better performance than the Davis → KIBA. The reason for this is the wide variety of protein types in the KIBA dataset. It causes, in Davis → KIBA, the approach faces compound–protein pairs that have not been observed in the training phase. Another finding is that our approach successfully predicts the selectivity of three compounds against five proteins of the PTP family. Also, CPI visualization with attention weights confirms that some interaction regions on proteins and compounds could be correctly captured by the proposed two-sided attention mechanism.

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